

Machine-learning projection of new-record generation hazard

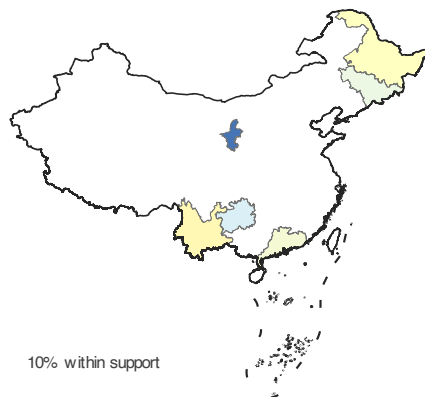
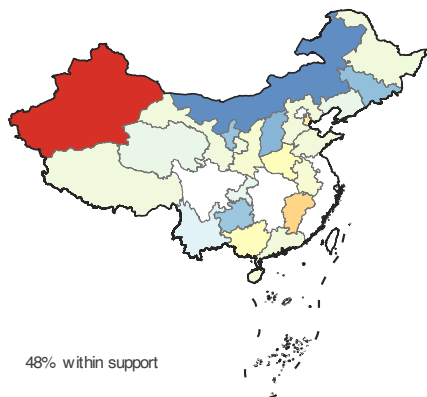
XGBoost on the same three predictors and identical future covariates; only rows inside the fitted covariate support

2030

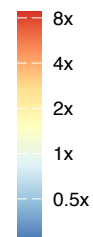
2050

2080

SSP2-4.5



Hazard relative to 2024



SSP5-8.5



Latent generation hazard relative to the 2024 baseline (log2 colour scale); province-by-year shocks set to expectation and reporting completeness set to 1. Base map GS(2019)1822. Gradient boosting returns boundary leaf values outside the training range, so its surface flattens rather than extrapolating.